Pulmonary Disorders II

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Learning Objectives

1. Design a treatment plan for a cystic fibrosis (CF) exacerbation.
2. Compare treatment strategies for pulmonary hypertension (PH) using the clinical classifications of PH.
3. Describe classifications and risk factors for increased severity of asthma exacerbation and treatment plans for patients with acute respiratory failure caused by asthma exacerbation.
4. Recognize evidence-based treatment options for acute exacerbations of chronic obstructive pulmonary disease (COPD).

Abbreviations in This Chapter

ABG	Arterial blood gas
AC/VC	Assist control/volume control
BNP	B-type natriuretic peptide
CF	Cystic fibrosis
CFTR	Cystic fibrosis transmembrane conductance regulator
COPD	Chronic obstructive pulmonary disease
CRP	C-reactive protein
CTEPH	Chronic thromboembolic pulmonary hypertension
CVP	Central venous pressure
ECOPD	Exacerbation of chronic obstructive pulmonary disease
ED	Emergency department
ET	Endothelin
ERA	Endothelin receptor antagonist
FEV_1	Forced expiratory volume in the first second of expiration
FIO_2	Fraction of inspired oxygen
ICU	Intensive care unit
MDI	Metered dose inhaler
mPAP	Mean pulmonary artery pressure
MV	Mechanical ventilation
$PaCO_2$	Partial pressure of carbon dioxide
PaO_2	Partial pressure of oxygen
PAH	Pulmonary artery hypertension
PCWP	Pulmonary capillary wedge pressure
PDE5i	Phosphodiesterase type 5 inhibitor
PEEP	Positive end-expiratory pressure
PH	Pulmonary hypertension
PVR	Pulmonary vascular resistance
RV	Right ventricular
SABA	Short-acting β-agonist
SaO_2	Oxygen saturation
SpO_2	Functional oxygen saturation
6MWD	6-minute walk distance

Self-Assessment Questions

Answers and explanations to these questions may be found at the end of this chapter.

1. A 21-year-old woman (height 62 inches, weight 50 kg) with a medical history significant for cystic fibrosis (CF) is admitted to the intensive care unit (ICU) with acute respiratory failure requiring mechanical ventilation (MV). After intubation, her arterial blood gas (ABG) results are as follows: pH 7.27, $PaCO_2$ 45 mm Hg, bicarbonate 22 mEq/L, PaO_2 55 mm Hg, and oxygen saturation (SaO_2) 88%. Her ventilator settings are as follows: assist control/volume control (AC/VC) mode, tidal volume 300 mL (6 mL/kg), respiratory rate 20 breaths/minute, fraction of inspired oxygen 60%, pressure support 5 cm H_2O, and positive end-expiratory pressure (PEEP) 5 cm H_2O. Her blood pressure is 110/70 mm Hg and heart rate is 95 beats/minute. Which is the best holistic therapy plan?
   A. Initiate cefepime 2 g intravenously every 8 hours and tobramycin 150 mg intravenously every 8 hours; intravenous fluid resuscitation to maintain a central venous pressure (CVP) goal of 10–14 mm Hg; dornase alfa and hypertonic saline 7% nebulization.
   B. Initiate cefepime 2 g intravenously every 8 hours and tobramycin 500 mg intravenously every 8 hours; intravenous fluid resuscitation to maintain a CVP goal of 10–14 mm Hg; dornase alfa and hypertonic saline 7% nebulization.
   C. Initiate cefepime 2 g intravenously every 8 hours and tobramycin 150 mg intravenously every 8 hours; intravenous fluid resuscitation to maintain a CVP goal of 10–14 mm Hg.
   D. Initiate cefepime 2 g intravenously every 8 hours and tobramycin 500 mg intravenously every 24 hours; diuresis to maintain a CVP goal of less than 4 mm Hg while mean arterial pressure is greater than 65 mm Hg; dornase alfa and hypertonic saline 7% nebulization.

2. Which adjunctive measure for a CF exacerbation has the highest quality of evidence for improving patient outcomes?
   A. Normal saline (0.9%) nebulization.
   B. Corticosteroid administration.
   C. Concurrent administration of intravenous and inhaled antibiotics.
   D. Aggressive chest physical therapy.

3. A 55-year-old woman with pulmonary artery hypertension (PAH) is admitted to the ICU for severe respiratory failure. She reports increased work of breathing for the past 5 days and full adherence to her PAH medication regimen, which includes macitentan 10 mg daily and sildenafil 40 mg three times daily. Her current vital signs are as follows: blood pressure 76/60 mm Hg, heart rate 140 beats/minute, respiratory rate 30 breaths/minute, and 85% Sao_2 on 6 L of nasal cannula. Right heart catheterization reveals the following: mean pulmonary artery pressure (mPAP) 50 mm Hg, right atrial pressure 25 mm Hg, cardiac index 1.9 L/minute/m^2, and pulmonary capillary wedge pressure (PCWP) 16 mm Hg. A transthoracic echocardiogram reveals an ejection fraction of 60% with severe right ventricular (RV) dilatation. Which regimen would be most appropriate for this patient?
   A. Dopamine infusion.
   B. Epoprostenol infusion.
   C. Phenylephrine infusion.
   D. Furosemide intravenous push.

4. Which best describes one of the primary treatment goals for the management of PAH?
   A. Achieve and maintain WHO functional class (FC) III or IV.
   B. Preserve 6-minute walk distance (6MWD) to greater than 300 m.
   C. Preserve RV size and function (right atrial pressure less than 8 mm Hg).
   D. Normalize B-type natriuretic peptide (BNP) (less than 200 ng/L).

5. A 42-year-old man presents to the emergency department (ED) with anxiety and shortness of breath. Auscultation reveals audible wheezing. He has difficulty speaking in full sentences. He has used his albuterol metered dose inhaler (MDI) at home for the past several hours without symptom resolution. His forced expiratory volume in the first second of expiration (FEV_1) is 35% of predicted. Which best classifies this patient's asthma exacerbation?
   A. Mild.
   B. Moderate.
   C. Severe.
   D. Life threatening.

6. A 55-year-old man presents to the ED with a 2-week history of progressive shortness of breath, wheezing, and dyspnea at rest. He is confused and speaking in short phrases but can indicate that his inhaled bronchodilator has failed to improve his symptoms for the past 2 days. He is afebrile with heart rate 116 beats/minute, respiratory rate 32 breaths/minute, and Sao_2 86% on room air. He is placed on supplemental oxygen and administered albuterol nebulization and methylprednisolone 125 mg intravenously × 1. One hour later, his oxygenation has not improved. Which adjunctive therapy would be best to add to his current regimen?
   A. Antimicrobial agents directed at community-acquired pathogens.
   B. Magnesium sulfate 2 g intravenously over 20 minutes.
   C. Theophylline 4.6 mg/kg intravenous loading dose; then obtain serum concentration in 30 minutes to guide further dosing.
   D. Acetylcysteine 20% 3 mL, nebulized every 6 hours.

7. A 66-year-old man presents to the ICU with acute respiratory failure from a chronic obstructive pulmonary disease exacerbation (ECOPD). He has had no exacerbations in the past 2 years. His home medications include albuterol hydrofluoroalkane 2 puffs four times daily as needed and tiotropium 1 puff once daily. The patient denies any recent sick contacts or changes in sputum production. He has no known drug allergies. He is placed on 3 L of nasal cannula (Sao_2 97%) and inhaled albuterol and ipratropium by nebulization. Which other therapy would be most appropriate for this patient?
   A. Azithromycin 500 mg intravenously daily.
   B. Prednisone 40 mg orally daily.
   C. Levofloxacin 500 mg orally daily.
   D. Methylprednisolone 125 mg intravenously every 6 hours.

8. A 65-year-old Black man (height 177.8 cm, weight 90 kg, SCr 1.5 mg/dL) presents with a medical history of end-stage renal disease on dialysis, COPD, hypertension, and hyperlipidemia. He is admitted from home to the ICU with respiratory failure. His symptoms before admission included shortness of breath, cough, increased sputum production, purulence, and fever. His respiratory viral panel is positive for rhinovirus. His Sao_2 is 85% on room air; he is placed on high-flow nasal cannula (HFNC) 40 L/minute, fraction of inspired oxygen (Fio_2) 60%. The patient's work of breathing decreases, and after 15 minutes, his Sao_2 is 97%. He is continued on the same HFNC settings. He has been administered methylprednisolone 60 mg intravenously once and azithromycin 500 mg intravenously once. Which best depicts the current intervention for this patient that has demonstrated an increase in mortality?
   A. Corticosteroid dose higher than prednisone 40 mg oral equivalent.
   B. Titration of oxygen to peripheral Sao_2.
   C. Azithromycin dose greater than 250 mg.
   D. Maintaining Sao_2 greater than 92%.

BPS Critical Care Pharmacy Examination Content Outline

This chapter covers the following sections of the Critical Care Pharmacy Examination Content Outline:

1. Domain 1: Critical Care
   a. Task A: 1-4
   b. Task B: 2, 3
2. Domain 2: Therapeutics and Patient Management
   a. Task A: 1–3, 6, 8
   b. Task B: 1, 2, 12–14
   c. Task C: 1, 4
3. Domain 3: Professional Practice
   a. Task A: 1
   b. Task B: 1

I. CYSTIC FIBROSIS

A. Cystic Fibrosis (CF) – Chronic disease process that affects many organs, including the pancreas, liver, and intestine, but primarily the lung (Chest 2004;125:1-39)
   1. CF is a recessive disorder caused by a mutation in the cystic fibrosis transmembrane conductance regulator (CFTR). CFTR controls the movement of chloride and bicarbonate across the cell membranes of many organs throughout the body (e.g., lungs, gastrointestinal tract, pancreas) (Chest 2018;154:383-93).
   2. Dysfunction of the CFTR within the airways also affects the ability of the airway membranes to absorb sodium and fluid through the epithelial sodium channel (ENaC), which creates a viscous mucus (Chest 2018;154:383-93). The pH of this mucus is more acidic, which impairs ciliary clearance and the function of phagocytic cells (Int J Mol Sci 2022;23:3513).
   3. Chronic therapy has been developed to target specific CFTR mutations, and such therapies may be seen in the ICU as home medications. Several of these agents have important drug-drug interactions as either substrates or inducers of CYP3A4 (Chest 2018;154:383-93).
   4. Social Determinants of Health in CF (Expert Rev Respir Med 2016;10:967-77)
      a. The term *socioeconomic status*, defined as access to education, material goods and services, social support, and position in society, contributes to differences in outcomes among patients with CF through varying exposure to allergens, quality of nutrition, chronic stress levels, and ability to self-manage disease, as well as access to medical care.
      b. Patients of a lower socioeconomic status have higher morbidity and mortality rates. Their disease progression related to lung function and nutrition are adversely affected as well.

B. Pulmonary Exacerbations of CF
   1. Definitions and significance
      a. There is no universally accepted definition for pulmonary exacerbations of CF (Respir Care 2020;65:233-51).
      b. Acute exacerbations include symptoms of increased cough, sputum production, shortness of breath, weight loss, and a decline in lung function.
      c. Pulmonary exacerbations of CF can have a significant impact on quality of life and disease progression. About 25% of patients with an exacerbation will not completely recover lung function at three months (Am J Respir Crit Care Med 2010;182:627-32).
   2. Common bacterial organisms (most common to least common)
      a. *Staphylococcus aureus* is the most common organism isolated from birth to teenage years.
      b. Methicillin resistant *Staphyloccocus aureus* (MRSA) becomes more common in the 11–30 years of age range.
      c. *Pseudomonas aeruginosa* has a prevalence rate of around 15% from birth to teenage years, but rapidly increases in prevalence after 17 years of age. Greater than 60% of adults with CF in the United States are infected with *P. aeruginosa*.
      d. *Haemophilus influenzae*
      e. *Stenotrophomonas maltophilia*
      f. *Achromobacter xylosoxidans*
      g. *Burkholderia cepacia complex*
      h. *Nontuberculous mycobacteria*
   3. Antibiotic therapy for pulmonary exacerbations of CF
      a. Antibiotic dosing is challenging because of the altered pharmacokinetics in patients with CF. Patients with CF can be distinguished by their large volume of distribution and augmented renal clearance.

b. Notably, most pulmonary exacerbations are not due to the acquisition of a new microorganism, and patients with CF commonly have routine cultures collected throughout their care. Using the patient's previous susceptibility pattern or restarting a previous successful antibiotic regimen is common practice.
   i. For patients with history of or suspected infection with *Pseudomonas aeruginosa*, initial treatment with two antipseudomonal agents is recommended, despite limited evidence that this practice improves outcomes (Am J Respir Crit Care Med 2009;180:802-8; Cochrane Database Syst Rev 2021;6:CD002007).
   ii. Two drugs from different classes are recommended rather than double β-lactams, with most data supporting a β-lactam in combination with an aminoglycoside, although toxicity risk and patient-specific factors should be taken into account.
   iii. Aggressive dosing of β-lactam antibiotics is recommended to optimize time above the minimum inhibitory concentration, primarily based on pharmacokinetic data. Current evidence is insufficient to recommend continuous infusion β-lactams for all patients with CF, but this practice should be considered on a case-by-case basis (Pharmacotherapy 2023;43:740-77; Respir Care 2009;54:522-37).
   iv. Extended-interval aminoglycoside dosing (e.g., once-daily dosing of tobramycin 10 mg/kg, targeting a peak concentration of 20–30 mg/L and a trough concentration of less than 1 mg/L) is as effective as conventional dosing, with possibly less nephrotoxicity (Lancet 2005;365:573-8; Cochrane Database Syst Rev 2017;3:CD002009).

c. Evidence is currently lacking for or against the simultaneous administration of inhaled and intravenous antibiotics for acute CF exacerbations (Table 1) (Am J Respir Crit Care Med 2009;180:802-8; Cochrane Database Syst Rev 2022;8:CD008319.

Table 1. Chronic Antimicrobial Therapy

Therapy	Clinical benefit(s)
Inhaled tobramycin[a]	• Improves lung function • Improves quality of life • Reduces exacerbations
Inhaled aztreonam[a]	• Improves lung function • Improves quality of life
Oral azithromycin[a]	• Improves lung function • Reduces exacerbations
Oral azithromycin[b]	• Reduces exacerbations

[a]Patients with history of *Pseudomonas aeruginosa*.

[b]Patients without history of *P. aeruginosa*.

Information from: Ann Am Thorac Soc 2014;11:1640-50

4. Duration of therapy
   a. A 10- to 14-day course of antibiotics is common practice, and durations are primarily driven by clinical response, either symptomatology or FEV_1 measurements.
   b. The Standardized Treatment of Pulmonary Exacerbation (STOP) trial reflected this precedent with a mean duration of intravenous antibiotics in adults of 16.2 days (J Cyst Fibros 2017;16:600-6).
   c. The STOP2 trial excluded ICU patients, but identified that for patients who improved within 7–10 days, 10 days of antibiotics was noninferior to 14 days. Patients without improvement should complete a 14-day course, but treating past 14 days was not associated with improved outcomes (Am J Respir Crit Care Med 2021;204:1295-305).

5. Exacerbation treatment outcomes
    a. Providers and practice will vary in defining when treatment is complete for pulmonary exacerbation. The STOP study identified the most common markers of success for providers including improvement in FEV_1 and patient-reported symptoms (J Cyst Fibros 2017;16:600-6).
    b. Though intravenous antibiotic therapy is known to improve lung function during an exacerbation, eradication of organisms in culture is neither required nor expected (Expert Opin Drug Metab Toxicol 2021;17:53-68).

C. Adjunctive therapy for prevention or treatment of pulmonary exacerbations of CF
1. Airway clearance techniques are a cornerstone of management of lung disease in CF.
    a. The type of therapy will be tailored to a patient's specific needs and preferences and the use of these interventions will likely increase during treatment of a pulmonary exacerbation (Respir Care 2009;54:522-37).
    b. High frequency chest wall oscillation, which involves an inflatable vest attached to a machine, is the most commonly used airway clearance technique (after infancy).
    c. Positive expiratory pressure/oscillating positive expiratory pressure involves breathing through a mask or mouthpiece to provide back pressure to the airways during expiration as gas also builds up behind the mucus by collateral ventilation (Cochrane Database Syst Rev 2019;11:CD003147).
    d. Despite a low level of evidence, physical exercise is recommended as a possible airway clearance therapy and is frequently used as either primary or secondary clearance for patients who are teenagers or older.
2. Nutrition
    a. Providing adequate nutrition according to patient-specific factors during acute exacerbations is key to maintaining metabolic function and promoting optimal outcomes.
    b. Administer pancreatic enzymes to treat exocrine dysfunction.
3. Continuation of patient's current chronic airway clearance therapy
    a. It is recommended to continue nebulized airway clearance therapies during an exacerbation (Table 2); there is limited data to describe the impact of this practice.

Table 2. Chronic Airway Clearance Therapy

Therapy	Mechanism	Clinical benefit(s)
Nebulized dornase alfa[a]	• Enzymatic cleavage of neutrophilic DNA • Reduces mucus viscosity	• Improves lung function • Improves quality of life • Reduces exacerbations
Nebulized hypertonic saline[a]	Osmotically active expectorant	
Nebulized mannitol[b]	Osmotically active expectorant	• Improves lung function

[a]Mogayzel PJ Jr, Naureckas ET, Robinson KA, et al. Cystic fibrosis guidelines: chronic medications for maintenance of lung health. Am J Respir Crit Care Med 2013;187:680-9.

[b]Flume PA, Amelina E, Daines CL, et al. Efficacy and safety of inhaled dry-powder mannitol in adults with cystic fibrosis: an international, randomized controlled study. J Cyst Fibros 2021;20:1003-9.

    b. The only airway clearance therapy that has been studied specifically during pulmonary exacerbations is hypertonic saline. This study of 132 patients, randomized to either 7% hypertonic saline or control, failed to detect a difference in the primary outcome of hospital length of stay, but did find patients in the treatment group were more likely to return to baseline FEV_1 and have significantly improved symptoms at discharge (Thorax 2016;71:141-7).
4. Data remains unclear on the role of continuing patient's chronic antimicrobial regimens during an exacerbation and practice will vary.

5. Corticosteroids
    a. Currently, there is insufficient evidence for administering corticosteroids (Am J Respir Crit Care Med 2009;180:802-8).
    b. However, current practice appears at odds with the 2009 guidelines. Though data from several small studies have been mixed, the STOP trial observed 21% of patients were treated with corticosteroids during a pulmonary exacerbation (J Cyst Fibros 2017;16:600-6).
    c. Attempts to clarify the role of corticosteroids are ongoing. The Prednisone in Cystic Fibrosis Pulmonary Exacerbations (PIPE) trial (currently recruiting) plans to compare 7 days of oral prednisone with placebo in patients with pulmonary exacerbation on intravenous antibiotics and who have not recovered their FEV_1 after 7 days of antibiotics. Study is still recruiting; currently unknown timeline for completion (NCT03070522).

Patient Case

Questions 1 and 2 pertain to the following case.
A 20-year-old woman is admitted to the ICU for an acute CF exacerbation. Before admission, her condition had been stable. She had maintained her ideal body weight and had just completed a home regimen of suppressive antibiotics. The patient requires MV for the management of hypoxic respiratory failure. She is initiated on AC/VC mode with a lung-protective strategy (4–8 mL/kg tidal volume).

1. Which of the following would be best to manage her acute CF exacerbation?
    A. Tobramycin nebulization; hypertonic saline 7% nebulization; and tube feedings to target a hypocaloric goal during her acute illness (15 kcal/kg/day).
    B. Ceftriaxone 2 g intravenously every 24 hours; tobramycin nebulization; normal saline 0.9% nebulization; and tube feedings to target her goal caloric intake (25 kcal/kg/day).
    C. Piperacillin/tazobactam 4.5 g intravenously every 6 hours; tobramycin 10 mg/kg intravenously once daily; hypertonic saline 7% nebulization; and tube feedings to target her goal caloric intake (25 kcal/kg/day).
    D. Cefepime 2 g intravenously every 8 hours extended infusion; tobramycin 10 mg/kg intravenously every 8 hours; hypertonic saline 7% nebulization; and tube feedings to target her goal caloric intake (25 kcal/kg/day).

2. After treatment with intravenous antibiotics, aggressive chest physical therapy, and her maintenance CF medications for 5 days, this patient improved and was extubated. She was moved out of the ICU on day 6 of her hospitalization. Which duration of antibiotics would be best to recommend for this patient?
    A. 7 days.
    B. 10 days.
    C. 14 days.
    D. Until cultures are negative.

II. PULMONARY HYPERTENSION

A. Pathophysiology and Definitions
   1. Pulmonary hypertension (PH) is a chronic, life-threatening disease characterized by vascular changes, including vasoconstriction, cellular proliferation, and thrombosis.
      a. Thromboxane A2 (potent vasoconstrictor) concentrations are increased.
      b. Prostacyclin (potent vasodilator, inhibitor of platelet aggregation, and antiproliferative properties) concentrations are decreased.
      c. Endothelin-1 (ET-1; potent vasoconstrictor, mitogenic properties on pulmonary artery smooth muscle cells) concentrations are increased.
      d. Nitric oxide concentrations (vasodilator, inhibitor of platelet activation, and inhibitor of vascular smooth muscle cell proliferation) are decreased.
   2. Clinical Classification of PH by the World Health Organization (WHO) (Chest 2019;155:565-86; Eur Heart J 2022;43:3618-731:
      a. Group 1 (pulmonary artery hypertension [PAH]): Idiopathic, heritable, or drug/toxin-induced PAH, PAH associated with various diseases, PAH with features of venous/capillary (PVOD/PCH) involvement, or persistent newborn PH
      b. Group 2: PH caused by left heart disease
      c. Group 3: PH caused by chronic lung disease and/or hypoxia
      d. Group 4: PH caused by chronic thromboembolic pulmonary hypertension (CTEPH) or other pulmonary artery obstructions
      e. Group 5: PH with unclear multifactorial mechanisms
   3. PH is defined as a mean pulmonary artery pressure (mPAP) greater than 20 mmHg at rest as measured by right heart catheterization (Eur Respir J 2019;53:1801913; Chest 2019;155:565-86).
      a. Hemodynamic definitions of PH (Eur Heart J 2022;43:3618-731)
         i. Groups 1, 3, 4, and 5: Precapillary PH defined as mPAP greater than 20 mmHg, pulmonary capillary wedge pressure (PCWP) less than or equal to 15 mm Hg, and pulmonary vascular resistance (PVR) greater than 2 Wood units
         ii. Groups 2 and 5: Isolated postcapillary PH defined as mPAP greater than 20 mmHg, PCWP greater than 15 mm Hg, and PVR less than or equal to 2 Wood units.
         iii. Groups 2 and 5: Combined pre- and postcapillary PH defined as mPAP greater than 20 mmHg, pulmonary capillary wedge pressure greater than 15 mm Hg, and PVR greater than 2 Wood units.

B. Diagnosis (Eur Heart J 2022;43:3618-731)
   1. Initial symptoms are nonspecific
      a. Common symptoms: Fatigue, shortness of breath with exertion, chest pain, and syncope
      b. Less common presentations: Dry cough, nausea, and vomiting
      c. Once PH diagnosis is confirmed, patient's WHO Functional Class will be assessed using these symptoms (Table 3).
   2. Due to the nonspecific pattern of symptoms, multiple tests will likely be needed to rule out other causes of such symptoms. This list is not complete, and additional tests may be performed based on patient-specific factors.
      a. Electrocardiogram
      b. Chest radiograph
      c. Pulmonary function tests
      d. Arterial blood gas
      e. Echocardiogram is used to predict likelihood of PH based on specific findings

i. Peak tricuspid regurgitation velocity (pTRV) is a key factor in suggesting the probability of PH, but it is only a suggestive factor and should not be used independently to predict PH. Guidelines differ in their recommendation for threshold of pTRV to suggest PH, with newer guidelines using > 2.8 m/second.
ii. Various measurements in the ventricles, pulmonary artery, inferior vena cava, and right atrium contribute to the determination of PH diagnosis.
iii. Echocardiogram should not be used independently for decision making about treatment for PH.

f. Ventilation/perfusion scan can be used to identify possible group 4 PH
g. Computed tomography (CT) or CT pulmonary angiography
h. Cardiac magnetic resonance imaging can provide noninvasive prognostic assessment of the right heart initially and at follow up.
i. Right heart catheterization is required to confirm diagnosis of PH
j. Perform vasoreactivity testing if appropriate

C. General management of PAH (group 1 PH)
1. Determine WHO Functional Class (FC): Defines patient's baseline functional status and is monitored throughout the disease to guide medication therapy. It is one of the primary predictors of survival throughout the patient's course of disease. (Table 3)

Table 3. WHO Functional Class Assessment[a]

Class	Definition
I	No symptoms (dyspnea, fatigue, syncope, chest pain) with normal activities
II	Symptoms with strenuous normal daily activities that slightly limit functional status and activity level; no symptoms at rest
III	Symptoms of dyspnea, fatigue, syncope, and chest pain with normal daily activities that severely limit functional status and activity level; no symptoms at rest
IV	Symptoms at rest; cannot perform normal daily activities without symptoms

[a]Humbert M, Kovacs G, Hoeper MM, et al. 2022 ESC/ERS Guidelines for the diagnosis and treatment of pulmonary hypertension. Eur Heart J 2022;43:3618-731.

2. Treatment goals for PH
a. Achieve and maintain WHO FC I or II.
b. Preserve 6-minute walk distance (6MWD) to greater than 440 m.
c. Preserve right ventricular (RV) size and function (right atrial pressure less than 8 mm Hg and cardiac index greater than or equal to 2.5 L/minute/m^2).
d. Normalize B-type natriuretic peptide (BNP) to less than 50 ng/L.
e. Maintain peak oxygen consumption greater than 15 mL/minute/kg and ventilatory equivalent for carbon dioxide less than 36 L/minute during cardiopulmonary exercise testing.
3. Supportive therapy
a. Physical activity and rehabilitation under the direction of a PH provider for patients in stable condition may improve quality of life and 6MWD
b. Oxygen: Maintain Sao_2 of 92% or greater and Pao_2 of 60 mm Hg or greater.
c. Diuretics should be used for the symptomatic management of RV dysfunction and signs of fluid overload; choice of diuretic is variable.
d. Digoxin increases cardiac output; consider in patients who develop atrial tachyarrhythmias although data specifically in PAH are not available.
e. Anticoagulation is not generally recommended for patients with PAH. Consider on an individualized basis if appropriate.

f. Aggressive treatment of sleep apnea and hypertension
g. Angiotensin-converting enzyme inhibitors, angiotensin-receptor blockers, angiotensin-receptor/neprilysin inhibitors, sodium-glucose cotransporter-2 inhibitors, β-blockers, or ivabradine are not recommended in patients with PAH unless indicated for a different comorbidity.
h. Treatment of iron deficiency anemia is indicated, defined by serum ferritin less than 100 mcg/L, or serum ferritin 100–299 mcg/L, and transferrin saturation less than 20%.
i. Patients with PAH should be vaccinated against influenza, *Streptococcus pneumoniae* and SARS-CoV-2 at minimum.

4. Special populations
   a. Pregnancy
      i. Despite advances in treatment and management, for patients with PAH who become pregnant, birthing parent and neonatal mortality rates are high.
      ii. People of childbearing potential should be provided with clear contraceptive recommendations. Some oral contraceptives have significant drug–drug interactions with PAH therapy, which should be considered and discussed with the patient.
   b. Surgery
      i. Surgical procedures pose a risk for patients with PAH. Risk increases with severity of PAH.
      ii. Decisions to perform surgery should be made by a multidisciplinary team familiar with the risks and benefits of both the procedure and PAH.

D. Specific treatment options for group 1 PAH
1. Vasodilator therapy with calcium channel blockers (CCBs) (diltiazem, amlodipine, nifedipine)
   a. Patients who should undergo vasoreactivity testing include those with idiopathic PAH, heritable PAH, and PAH secondary to toxins or drugs.
   b. Considered first-line treatment for patients who have a positive response to acute vasoreactivity testing (reduced mPAP by 10 mm Hg or more to an mPAP of 40 mm Hg or less with increased or unchanged cardiac output) (N Engl J Med 1992;327:76-81)
   c. CCBs improve mPAP, PVR, WHO FC, and survival in patients with PAH and a positive vasoreactivity test (Circulation 1987;76:135-41; N Engl J Med 1992;327:76-81).
   d. Patients are not considered candidates for CCB therapy if they have RV dysfunction, depressed cardiac output, or WHO FC IV symptoms.
   e. *Marked hemodynamic improvement* in patients on CCB therapy is defined as mPAP greater than 30 mm Hg and a PVR less than 4 Wood units. If patients do not reach this threshold of response, CCB therapy can be continued or stopped according to patient-specific factors; some patients may experience deterioration when CCBs are withdrawn.
2. Treatment-naïve patients with PAH (nonresponders to vasoreactivity testing)
   a. Drug therapy should be evaluated on the basis of the patient-specific factors below (Chest 2019;155:565-86; Eur Heart J 2022;43:3618-731):
      i. Mortality risk assessment using the 3-strata model (low, intermediate, high), as shown in Table 4
      ii. Presence of cardiopulmonary comorbidities that are associated with increased risk of LV diastolic dysfunction (obesity, hypertension, diabetes, coronary heart disease)
      iii. Tolerance of combination therapy
      iv. Ability and willingness to manage parenteral prostacyclins
      v. Access to resources
   b. Initial therapy in patients without cardiopulmonary comorbidities
      i. Low- or intermediate-risk strata (Table 4): combination therapy with an endothelin receptor antagonist (ERA) and a phosphodiesterase type 5 inhibitor (PDE5i)
      ii. High-risk strata: combination therapy with ERA, PDE5i, and parenteral prostacyclin

Table 4. Three-Strata Risk Assessment for Patients with Pulmonary Artery Hypertension

Multivariable Assessment of Mortality Risk to Guide Initial Therapy			
Assessment	**3-Risk Strata**		
Estimation of 1-year mortality	**Low Risk < 5%**	**Intermediate Risk 5%–20%**	**High Risk > 20%**
Signs of right heart failure	No	No	Yes
Progression of symptoms and clinical manifestations	No	Slow	Rapid
Syncope	No	Occasional during strenuous exercise	Repeated with little or normal physical activity
WHO-FC	I, II	III	IV
6-Minute walk distance	> 440 m	165–440 m	< 165 m
Cardiopulmonary exercise testing	Peak VO_2 > 15 mL/min/kg (> 65% predicted) VE/VCO_2 slope < 36	Peak VO_2 11–15 mL/min/kg (35%–65% predicted) VE/VCO_2 slope 36–44	Peak VO_2 < 11 mL/min/kg (< 35% predicted) VE/VCO_2 slope > 44
BNP or NT-proBNP	BNP < 50 ng/L NT-proBNP < 300 ng/L	BNP 50–800 ng/L NT-proBNP 300–1100 ng/L	NP > 800 ng/L NT-proBNP > 1100 ng/L
Echocardiography parameters	RA area < 18 cm^2 TAPSE/sPAP > 0.32 mm/mm Hg No pericardial effusion	RA area 18–26 cm^2 TAPSE/sPAP 0.19–0.32 mm/mmHg Minimal pericardial effusion	RA area > 26 cm^2 TAPSE/sPAP < 0.19 mm/mm Hg Moderate/large pericardial effusion
Cardiac magnetic resonance imaging	RVEF >54% SVI >40 mL/m^2 RVESVI <42 mL/m^2	RVEF 37–54% SVI 26–40 mL/m^2 RVESVI 42–54 mL/m^2	RVEF < 37% SVI < 26 mL/m^2 RVESVI > 54 mL/m^2
Hemodynamics	RAP < 8 mm Hg CI ≥ 2.5 L/min/m^2 SVI > 38 mL/m^2 SvO_2 > 65%	RAP 8–14 mm Hg CI 2.0–2.4 L/min/m^2 SVI 31–38 mL/m^2 SvO_2 60–65%	RAP >14 mm Hg CI < 2.0 L/min/m^2 SVI < 31 mL/m^2 SvO_2 < 60%

BNP, B-type natriuretic peptide; CI, cardiac index; NT-proBNP, N-terminal pro-brain natriuretic peptide; RA, right atrium; RAP, right atrial pressure; sPAP, systolic pulmonary artery pressure; SvO_2, mixed venous oxygen saturation; RVESVI, right ventricular end-systolic volume index; RVEF, right ventricular ejection fraction; SVI, stroke volume index; TAPSE, tricuspid annular plane systolic excursion; VE/VCO_2, ventilatory equivalents for carbon dioxide; VO_2, oxygen uptake; WHO-FC, World Health Organization functional class.

Adapted from Humbert M, Kovacs G, Hoeper MM, et al. 2022 ESC/ERS guidelines for the diagnosis and treatment of pulmonary hypertension. Eur Heart J 2022;43:3618-731.

c. Initial and subsequent therapy in patients with cardiopulmonary comorbidities: because these patients are under-represented in clinical studies, no evidence-based recommendations can be provided. Although these patients are often treated similarly in practice, they have more treatment-related adverse effects or tolerability issues with combination therapy.
   i. All risk strata: oral monotherapy with PDE5i or ERA
   ii. Follow-up therapeutic changes should be based on patient-specific characteristics

3. Subsequent therapeutic options are based on the four strata risk assessment (Table 5). All 3 variables are recommended to be included in each follow-up assessment for better accuracy at predicting mortality. If only 2 variables are available, it is preferred to include 1 biomarker (BNP or NT-proBNP) and 1 functional marker (either WHO-FC or 6MWD).
   a. Low-risk strata: Continue current therapy
   b. Intermediate/Low-risk strata
      i. Add prostacyclin receptor antagonist (PRA)—or—
      ii. Switch from PDE5i to soluble guanylate cyclase stimulator (sGCs)

Table 5. Four-Strata Risk Assessment For Patients With Pulmonary Artery Hypertension

Multivariable Assessment of Mortality Risk to Guide Follow-Up Therapy				
Assessment	**4-Risk Strata**			
Prognostic determinants	**Low**	**Intermediate/Low**	**Intermediate/High**	**High**
Points[a]	1	2	3	4
WHO-FC	I or II	—	III	IV
6-Minute walk distance	> 440 m	320–440 m	165–319 m	< 165 m
BNP or NT-proBNP	< 50 ng/L < 300 ng/L	50–199 ng/L 300–649 ng/L	200–800 ng/L 650–1100 ng/L	> 800 ng/L > 1100 ng/L

[a]Divide the sum of all points by the number of variables and round to the next whole number.

BNP, B-type natriuretic peptide; NT-proBNP, N-terminal pro-brain natriuretic peptide; WHO-FC, World Health Organization functional class.

Adapted from Humbert M, Kovacs G, Hoeper MM, et al. 2022 ESC/ERS guidelines for the diagnosis and treatment of pulmonary hypertension. Eur Heart J 2022;43:3618-731.

Table 6. Endothelin Receptor Antagonists (oral)

	Bosentan	**Ambrisentan**	**Macitentan**
Receptor affinity	Blocks ET_a and ET_b	Preferentially blocks ET_a	Blocks ET_a and ET_b
Half-life (hr)	5	9	16
Approved dose	62.5–125 mg BID	5–10 mg daily	10 mg daily
Outcomes	Delay clinical worsening, improves exercise capacity, improves WHO-FC, and hemodynamics	Improved symptoms, exercise capacity, hemodynamics, and time to clinical worsening	Increased exercise capacity, improved a composite end point of clinical worsening
Adverse effects	Hepatotoxicity (except ambrisentan), peripheral edema, anemia, contraindicated during pregnancy (REMS programs)		
Drug interactions	Glyburide (contraindicated) and cyclosporine (contraindicated), CYP2C9 and CYP3A4 inhibitors and inducers, hormonal contraceptives, warfarin, sildenafil and tadalafil	Caution with cyclosporine (max ambrisentan dose is 5 mg daily)	CYP2C19 and strong CYP3A4 inhibitors and inducers

BID = twice daily; ET = endothelin; REMS = risk evaluation and mitigation strategies.

Information from: Galiè N, Barberà JA, Frost AE, et al. Initial use of ambrisentan plus tadalafil in pulmonary arterial hypertension. N Engl J Med 2015;373:834-44; Humbert M, Kovacs G, Hoeper MM, et al. 2022 ESC/ERS Guidelines for the diagnosis and treatment of pulmonary hypertension. Eur Heart J 2022;43:3618-731; Pulido T, Adzerikho I, Channick RN, et al. Macitentan and morbidity and mortality in pulmonary arterial hypertension. N Engl J Med 2013;369:809-18.

c. Intermediate/High- or high-risk strata
   i. Add a parenteral prostacyclin—and/or—
   ii. Evaluate for lung transplant

E. Drug Information by Class
1. Endothelin receptor antagonists (ERAs) (Table 6)
   a. Competitively block the action of ET-1 from acting on two types of ET receptors: ET_a and ET_b. This blockade prevents endogenous ET-1 from its normal actions:
      i. ET_a = vasoconstriction and cell proliferation
      ii. ET_b = vasodilation and antiproliferation of pulmonary endothelial cells
   b. Minimal place in therapy for initiation in critically ill patients
2. PDE-5 inhibitors (Table 7)
   a. Inhibit PDE-5, which prevents the breakdown of cyclic guanosine monophosphate in pulmonary vascular smooth muscle. Increased cyclic guanosine monophosphate potentiates pulmonary vascular smooth muscle and vasodilation of the pulmonary vascular bed.
   b. Indicated for patients who continue to be symptomatic on intravenous epoprostenol.
   c. Intravenous formulation of sildenafil is available for patients who temporarily cannot ingest tablets; however, this formulation is limited by its hemodynamic effects.

Table 7. PDE5i

	Sildenafil (Revatio, Liqrev)	**Sildenafil (Revatio)**	**Tadalafil (Adcirca, Alyq, Tadliq)**
Half-life (hr)	4		35
Approved dose	20 mg oral TID	10 mg IV TID	40 mg oral daily; dose adjustment necessary for renal impairment
Outcomes	WHO FC II and III: Improve 6MWD[a] WHO FC III: Improve WHO FC[a]		WHO FC II and III: Improve 6MWD[b]
Outcomes	WHO FC III: Improve WHO FC[a]	WHO FC II and III: Improve 6MWD[b]	WHO FC II and III: Delay clinical worsening[a] WHO FC III: Improve WHO FC
Adverse effects	Headache, epistaxis, flushing, dyspepsia, hypotension, visual alterations		
Drug interactions	Contraindicated in patients receiving nitrates; avoid with strong CYP3A4 inhibitors and inducers; contraindicated with riociguat		

[a]Klinger JR, Elliott CG, Levine DJ, et al. Therapy for pulmonary arterial hypertension in adults. Chest 2019;155:565-86.

[b]Galiè N, Barberà JA, Frost AE, et al. Initial use of ambrisentan plus tadalafil in pulmonary arterial hypertension. N Engl J Med 2015;373:834-44.

IV = intravenously; TID = three times daily; WHO-FC = World Health Organization functional class; 6MWD = 6-min walk distance.

3. Soluble guanylate cyclase stimulators
   a. Riociguat exerts its effect through two mechanisms: Sensitizes endogenous soluble guanylate cyclase by stabilizing nitric oxide–soluble guanylate cyclase bonding and directly stimulates soluble guanylate cyclase through an auxiliary binding site. Both processes increase cyclic guanosine monophosphate, which influences vascular tone, proliferation, fibrosis, and inflammation.
   b. For patients on ERA/PDE5i therapy and at low-intermediate risk of death, switching from PDE5i to riociguat could be considered (Lancet Respir Med 2021:9:573-84).
   c. Elimination half-life is 12 hours. If cannot be administered for 3 days or more, retitration is required; however, a PH specialist should always be involved for issues with access and reinitiation.
   d. Approved dose is 1–2.5 mg three times daily (higher doses for smokers). In patients at risk of systemic hypotension, a starting dose of 0.5 mg three times daily may be used (2022 guidelines)

Table 8. Additional Roles in Therapy for Riociguat

Failed monotherapy	Add	Clinical outcomes	WHO FC III-IV
Bosentan	Riociguat	Improve 6MWD Improve WHO FC Delay time to clinical worsening	Recommended[a]
Ambrisentan			Recommended[a]
Prostacyclins (inhaled)			Recommended[a]

[a]Consensus-based level of evidence (Chest 2019;155:565-86)

WHO FC = World Health Organization functional class; 6MWD = 6-min walk distance.

e. Adverse effects include hypotension, hemoptysis, headache, dizziness, dyspepsia, nausea, diarrhea, vomiting, and anemia.
f. Contraindicated in patients receiving nitrates and phosphodiesterase inhibitors because of hypotension; avoid with strong CYP3A4/2C8 inhibitors and inducers and with P-glycoprotein/ breast cancer resistance protein inhibitors
g. Riociguat risk evaluation and mitigation strategies (REMS) program for embryo-fetal toxicity

4. Prostacyclins
   a. Parenteral prostacyclins (Table 9–10)
      i. Mimic the actions of PGI_2 (endogenous prostacyclin): Direct vasodilation of pulmonary and systemic arterial vascular beds, inhibition of platelet aggregation, and antiproliferative effects
      ii. Continuous-infusion epoprostenol is the most thoroughly studied medication approved for treating PAH and has been shown to prolong survival.
         • 100% survival with intravenous epoprostenol compared with 80% survival with conventional therapies at 12 weeks (p=0.003) (N Engl J Med 1996;334:296-301)
         • 88% survival with intravenous epoprostenol compared with 59% expected survival in data from historical controls at 1 year (p<0.001) (Circulation 2002;106:1477-82)
         • 55% survival with intravenous epoprostenol compared with 28% survival in historical controls at 5 years (p<0.0001) (J Am Coll Cardiol 2002;40:780-8)
      iii. Complications related to delivery include the need for a dedicated intravenous line, local catheter/bloodstream infections, and catheter-related thrombosis.
   b. Inhaled prostacyclins (Table 9 and Table 10) – Advantage over intravenous route because of selective pulmonary vasodilation with minimal systemic effects

Table 9. Additional Roles in Therapy for Prostacyclins

Failed monotherapy	Add	Clinical outcomes
1–2 classes of oral agents	IV epoprostenol	Improve WHO FC Improve 6MWD
1–2 classes of oral agents	IV treprostinil	Improve 6MWD
IV epoprostenol	Sildenafil (2022 guidelines)	Improve 6MWD
ERA	Inhaled treprostinil	Improve 6MWD
	Inhaled iloprost	Improve WHO FC Delay clinical worsening
PDE5i	Inhaled treprostinil	Improve 6MWD
	Inhaled iloprost	Improve WHO FC Delay clinical worsening

6MWD = 6-min walk distance.

Information from: Klinger JR,, Elliott CG, Levine DJ, et al. Therapy for pulmonary arterial hypertension in adults. Chest 2019;155:565-86; Humbert M, Kovacs G, Hoeper MM, et al. 2022 ESC/ERS guidelines for the diagnosis and treatment of pulmonary hypertension. Eur Heart J 2022;43:3618-731.

Table 10. Prostacyclins

	Epoprostenol		Treprostinil	Treprostinil	Iloprost	Treprostinil
Brand name	**Flolan**	**Veletri**	**Remodulin**	**Tyvaso**	**Ventavis**	**Orenitram**
Route of administration	IV		SC and IV	Inhalation	Inhalation	Oral
Initial dose	1–4 ng/kg/min IV		1.25 ng/kg/min IV or SC; dose adjustments in hepatic impairment	18 mcg four times daily (solution) 16 mcg four times daily (dry powder)	2.5–5 mcg six to nine times daily	0.25 mg every 12 hr or 0.125 mg every 8 hr; caution with strong CYP2C8 inhibitors and severe; contraindicated in severe hepatic impairment
Half-life	6 min		4 hr		25 min	4 hr
Adverse effects	Flushing, headache, diarrhea nausea/ vomiting, jaw pain, thrombocytopenia			Cough, throat irritation, bronchospasm, hypotension		Headache, diarrhea, nausea, flushing, jaw pain, and abdominal discomfort
Cautions	Rebound PH with abrupt discontinuation; dosing weight should not be changed without consultation with PH specialist					

CYP = cytochrome P450; IV = intravenous; PH = pulmonary hypertension; SC = subcutaneous.

Information from: Humbert M, Kovacs G, Hoeper MM, et al. 2022 ESC/ERS Guidelines for the diagnosis and treatment of pulmonary hypertension. Eur Heart J 2022;43:3618-731.

c. Oral treprostinil (Table 10):
   i. FDA approved as monotherapy for PAH, improved 6MWD
   ii. The addition of oral tresprostinil to ERA or PDE5i/riociguat monotherapy is recommended to reduce the risk of morbidity and mortality in patients with idiopathic, heritable, or drug-associated PAH (Eur Heart J 2022;43:3618-731).

5. Selective prostacyclin receptor (IP receptor) agonist
   a. Selexipag is an oral nonprostanoid that targets the prostacyclin pathway.
   b. The 2019 CHEST guideline update changed their recommendation for this agent based on the included studies not achieving their prespecified definition of clinically significant improvement, defined as 33 m in the 6MWD. However, practitioners still use selexipag despite these guidelines not providing a recommendation for or against the use of this agent.
   c. The 2022 ESC/ERS guidelines support
      i. Consider the addition of selexipag to ERA/PDE5i therapy for patients who present with low/intermediate risk of death at follow-up
      ii. Add selexipag if unable to use a parenteral prostacyclin for patients at intermediate/high or high risk.
      iii. Add selexipag to ERAs and/or PDE5is to reduce the risk of morbidity and mortality events.
   d. It is not recommended to start macitentan, tadalafil, and selexipag as initial therapy, given the lack of current evidence for this strategy (J Am Coll Cardiol 2021;78:1393-403).
   e. Elimination half-life is 0.8–3 hours and, for the active metabolite, 6–14 hours. If unable to administer for 3 days or more, retitration is required; however, a PH specialist should always be involved for issues with access and reinitiation.
   f. Approved dose is 200 mcg twice daily; increase at weekly intervals to the highest tolerated dose (maximum dose 1600 mcg twice daily); adjustment necessary for moderate hepatic impairment

g. Dosing should be reduced to once daily in patients receiving moderate CYP2C8 inhibitors whereas selexipag is contraindicated in patients receiving strong CYP2C8 inhibitors (e.g., gemfibrozil); dose increase necessary with CYP2C8 inducers
h. Adverse effects include headache, diarrhea, nausea and jaw pain.

F. Managing decompensated PH
1. Decompensated PH generally presents with worsening symptoms of right heart failure such as shortness of breath, syncope, fluid retention, or even multiorgan failure due to poor perfusion.
2. Markers of imminent mortality from PH/severe right sided heart failure include a decline in central venous oxygen saturation, lactic acidosis, and decreased urine output.
3. Manage contributing factors such as infections, anemia, arrhythmias, rebound PH (nonadherence or ineffective dosing), hypoxemia, acidosis, and metabolic abnormalities.
4. Supportive therapies include optimizing RV preload through fluid optimization (diuretics or dialysis), maintaining aortic root pressure, improving RV contractility, and reducing RV afterload.
5. Temporary discontinuation of CCBs and β-blockers to eliminate negative inotropic effects
6. Hemodynamic support (Table 11)
   a. Maintaining aortic root pressure and minimizing RV ischemia can be accomplished using vasopressors, which increase the systemic vascular resistance and ultimately improve RV perfusion (Crit Care Med 2007;35:2037-50).
   b. Direct effects on the pulmonary circulation from vasopressors may increase the PVR, potentially leading to further clinical decompensation.
   c. Few studies have been published to help guide selection of the optimal vasopressor in patients with PH; recommendations are extrapolations from other patient populations.

Table 11. Agents Used for Hemodynamic Support for Decompensated PH

Drug	Site of Action (receptor activity)	Effects on PVR	Effects on CO	Comments
Dopamine	Dose-dependent dopaminergic, α_1 and β_1	↑ or ↔	↑	May not improve RV ejection fraction; arrhythmias
Norepinephrine[a]	$\alpha_1 > \beta_1$	↑	↑ or ↔	Decreased mortality in subgroup of cardiogenic shock and decreased rate of arrhythmias compared with dopamine in a randomized trial
Phenylephrine	α_1	↑	↔	Reflex bradycardia may be harmful in the setting of RV failure
Epinephrine	$\alpha_1, \beta_1 > \beta_2$	↑ or ↔	↑	Arrhythmias, hyperglycemia, increased lactate concentrations
Vasopressin	V_1 receptors	↑ or ↔	↑ or ↓	Use low dose (≤ 0.03 units/min)
Dobutamine	$\beta_1 > \beta_2$	↓	↑	Combine with peripheral vasoconstrictor to attenuate systemic vasodilation
Milrinone	PDE-3 inhibitor	↓	↑	Combine with peripheral vasoconstrictor to attenuate systemic vasodilation

[a]De Backer D, Biston P, Devriendt J, et al. Comparison of dopamine and norepinephrine in the treatment of shock. N Engl J Med 2010;362:779-89.

CO = cardiac output; PDE-3 = phosphodiesterase-3; PVR = pulmonary vascular resistance; RV = right ventricular.

   d. Inotropes are used to further augment the cardiac output of the RV and may improve PVR. Because of systemic vasodilatory properties from inotropes, expect possible systemic hypotension and need for vasopressors.
7. Unloading the RV with pulmonary vasodilators is essential to controlling decompensated PH and RV failure.
8. If a patient is not responsive to therapy and may be a transplant candidate, extracorporeal life support can be considered as a bridge to lung transplant. Candidates must be chosen prudently because RV failure must be reversible for this strategy to be successful.

Patient Case

Questions 3 and 4 pertain to the following case.
A 44-year-old man is transferred to the medical ICU for treatment of his worsening PAH. He currently receives no PAH treatment. The patient has had increased dyspnea on exertion for the past 6 months with exercise. For the past 2 months, he has had difficulty going for walks with his dog without getting fatigued, and occasionally, he has chest pain. In the past few days, he has had severe shortness of breath at rest, which prompted him to come to the hospital. His physical examination is remarkable for blood pressure 105/64 mm Hg and heart rate 85 beats/minute. Lung examination is clear, and extremities are notable for trace edema. An echocardiogram reveals an elevated pulmonary systolic pressure and a normal ejection fraction. The patient has an unfavorable response to vasodilator challenge. Pertinent laboratory data are serum urea nitrogen 10 mg/dL, SCr 0.6 mg/dL, AST 160 U/L, and ALT 100 U/L.

3. Which best describes his WHO FC and risk of mortality in the next year?
   E. WHO FC III, mortality risk 5%.
   F. WHO FC III, mortality risk 10%.
   G. WHO FC IV, mortality risk 15%.
   H. WHO FC IV, mortality risk greater than 20%.

4. Which medication therapy would be best to recommend initiating at this time?
   A. Epoprostenol infusion at 2 ng/kg/minute.
   B. Diltiazem 180 mg orally daily.
   C. Macitentan 10 mg orally daily.
   D. Sildenafil 10 mg intravenously three times daily.

III. ASTHMA EXACERBATION

A. Pathophysiology and Classification of Exacerbations
   1. Asthma is characterized by variable and recurring symptoms, airflow obstruction, bronchial hyperresponsiveness, and underlying inflammation (NAEPP 2007).
   2. Asthma exacerbations are, also termed *acute severe asthma*, characterized by decreases in expiratory airflow that can be quantified by spirometry or peak expiratory flow.
   3. Definitions and classifications of asthma exacerbations vary between experts and guidelines. Both terms below are considered life threatening exacerbations.

a. Status asthmaticus is an acute severe asthma exacerbation that does not respond to initial intensive therapy, with the potential for pulmonary compromise and death.
b. Near-fatal asthma is status asthmaticus that progresses to respiratory failure.

Table 12. Classification of Asthma Exacerbations in Acute Care Facility (GINA 2023)

	Symptoms	Initial PEF (or FEV_1)	Clinical Course
Mild/ Moderate	• Talks in phrases • Prefers sitting to laying down • Not agitated • RR increased • Accessory muscles not used • Heart rate 100–120 beats/min • SaO_2 90%–95%	> 50% of predicted or personal best	• Usually cared for at home/primary care setting • Relief from frequent inhaled SABA • Consider addition of ipratropium • Recommended course of oral CS • SaO_2 goal 93%–95%
Severe	• Talks in words • Sits hunched forward • Agitated • RR > 30 breaths/min • Accessory muscles used • Heart rate > 120 beats/min • SaO_2 < 90%	≤ 50% of predicted or personal best	• Requires ED visit and likely hospitalization • Recommended inhaled SABA + ipratropium bromide • SaO_2 goal 93%–95% • Oral or IV corticosteroids • Consider IV magnesium • Consider high dose ICS
Life threatening	• Drowsy • Confused • Silent chest (absent breath sounds)	No definition	• Requires hospitalization; possible ICU • Minimal or no relief from frequent inhaled SABA • IV corticosteroids • Adjunctive therapies are helpful

ED = emergency department; FEV_1 = forced expiratory volume in the first second of expiration; ICS = inhaled corticosteroid; ICU = intensive care unit; IV = intravenous; PEF = peak expiratory flow; RR = respiratory rate; SABA = short-acting β-agonist; SaO_2 = oxygen saturation.

Adapted from: Venkatesan P. 2023 GINA report for asthma. Lancet Respir Med 2023;11:589.

B. Common Causes of Asthma Exacerbations
   1. Viral infections
   2. Allergen exposure
   3. Food allergy
   4. Air pollution
   5. Seasonal weather changes
   6. Poor adherence to inhaled corticosteroids

C. Mortality Risk Factors
   1. History of near-fatal asthma (e.g., requiring MV)
   2. Hospitalization or ED visit for asthma in the past year
   3. Active use of oral corticosteroids or completion of recent course for asthma
   4. Not currently using inhaled corticosteroids
   5. Use of more than one canister of short-acting β-agonists (SABAs) per month
   6. Poor adherence to inhaled corticosteroid containing asthma medications and/or poor adherence to asthma action plan

7. Social history that includes major psychosocial problems or psychiatric illness
8. Food allergies
9. Comorbidities including pneumonia, diabetes, and arrhythmias

D. Social Determinants of Health in patients with asthma (Ann Allergy Asthma Immunol 2022;128:5-11)
   1. Structural racism and inequities in socioeconomic status have had profound effects on both the prevalence of asthma and frequency of exacerbations in Black and Latinx patients.
   2. Adults with a household income < $50,000 had a 1.6 times higher rate of asthma treatment failure and 2 times higher rate of asthma exacerbations compared with those with a household income >=$50,000.
   3. Long-term stress, healthcare access, environmental factors, education level, food accessibility and other factors can impact both disease prevalence and management in patients with asthma. The full extent of which of these variables has the highest effects is still under investigation.

E. COVID-19 (SARS-CoV-2) and Asthma (GINA 2023)
   1. Patients with well-controlled asthma do not appear to be at increased risk of infection or mortality due to SARS-CoV-2.
   2. However, patients who recently required oral corticosteroids for their asthma and those hospitalized with severe asthma were at increased risk of death from COVID-19.
   3. Patients on inhaled corticosteroids for their asthma had lower mortality from COVID-19 compared with patients without underlying lung disease (Lancet Respir Med 2021;9:699-711).
   4. Both GINA and the Centers for Disease Control and Prevention recommend that patients receive vaccination against COVID-19 including boosters if applicable.
   5. If patients are receiving biologics for their severe asthma, it is not recommended to give the biologic and COVID-19 vaccine on the same day.

F. Alternative Causes (mimic asthma exacerbation)
   1. Upper airway: Vocal cord dysfunction, anaphylaxis, laryngeal stenosis
   2. Central airway: Tracheomalacia, tracheal stenosis mucus plugging
   3. Lower airway: Bronchiolitis, COPD, valvular heart disease, diastolic heart dysfunction

G. ABG Assessment
   1. Patients with acute severe asthma initially experience a respiratory alkalosis.
   2. As respiratory status worsens, arterial carbon dioxide increases (patient exhaustion, inadequate alveolar ventilation and/or an increase in physiologic dead space), leading to respiratory acidosis.
   3. Metabolic (lactic) acidosis may coexist. Lactate production presumably stems from the use of high-dose β-agonists, increased work of breathing resulting in anaerobic metabolism of the ventilatory muscles, and tissue hypoxia.

H. Oxygen
   1. Oxygen therapy is important in managing acute severe asthma.
   2. Oxygen by nasal cannula or mask should be administered to patients with severe exacerbations with hypoxemia to achieve Sao_2 values greater than 93%–95% (pregnant women and patients with a cardiac history may require higher goals) (GINA 2023; Thorax 2011;66:937-41).

I. Noninvasive Ventilation: Data are insufficient for severe exacerbations. If attempted, patients should not be sedated to tolerate noninvasive ventilation.

J. Mechanical Ventilation
   1. Indications
      a. Worsening hypoxemia or hypercarbia
      b. Drowsiness or altered mental status
      c. Hemodynamic instability
      d. Increased work of breathing
   2. Low minute ventilation (by reduced tidal volume and/or respiratory rate), high inspiratory flow rate, and minimal PEEP on the ventilator will help minimize dynamic hyperinflation.

K. β-Agonists
   1. SABAs stimulate the β_2-receptors on smooth muscle cells, leading to relaxation of respiratory smooth muscle and causing bronchodilation and decreased airway obstruction.
   2. SABAs are the cornerstone in managing acute severe asthma.
   3. Patients with an asthma exacerbation should receive a SABA repeatedly by either an MDI or nebulization at presentation, and the SABA should be continued until acute symptoms have resolved (NAEPP 2007).
   4. Patients with acute severe exacerbations may benefit from continuous versus intermittent nebulization of SABAs (Cochrane Database Syst Rev 2003;4:CD001115).
      a. Intermittent dosing of albuterol: 2.5–5 mg every 20 minutes for three doses; then 2.5–10 mg every 1–4 hours as needed
      b. Continuous nebulization of albuterol: 10–15 mg/hour
   5. Systemic β-agonists can be considered if the patient does not respond to inhaled therapy after several hours (Chest 2022;162:747-56).
      a. These agents have no proven advantage over inhaled agents or over each other and are not recommended for routine use because of the high rate of adverse effects.
      b. However, if the patient's acute asthma exacerbation is caused by anaphylaxis or angioedema, epinephrine would be recommended.
      c. Epinephrine 1 mg/mL 0.3–0.5 mg intramuscularly/subcutaneously every 20 minutes up to three doses
      d. Epinephrine infusion 0.1 mcg/kg/minute
      e. Terbutaline 0.25 mg subcutaneously every 20 minutes for three doses
   6. Adverse effects include tremor, tachyarrhythmias, hypokalemia, tachyphylaxis, hyperglycemia, and type B lactic acidosis.

L. Anticholinergic Agents
   1. Inhaled anticholinergic agents selectively bind to the muscarinic receptors on smooth muscle cells in the airways and thereby reduce bronchoconstriction.
   2. Short-acting inhaled anticholinergic agents (ipratropium bromide) should be given in combination with a SABA to promote additional bronchodilation through a different pathway.
   3. Adding ipratropium to inhaled albuterol compared with using albuterol alone in patients with severe asthma improved the response; however, outcomes with this combination in status asthmaticus or near-fatal asthma remain elusive (Am J Respir Crit Care Med 2000;161:1862-8). Combination therapy consisting of inhaled short-acting anticholinergics and a SABA compared with a SABA alone has reduced hospitalization rates in severe asthma exacerbations. However, combination therapy may increase the risk of adverse drug events, including agitation and palpitations (Cochrane Database Syst Rev 2017;1:CD001284).
   4. Adverse effects include headache, flushed skin, blurred vision, tachycardia, palpitations, and urinary retention.

M. Corticosteroid Therapy
   1. Corticosteroids decrease airway obstruction during an asthma exacerbation by decreasing inflammation, increasing the number of β2-receptors and increasing their responsiveness to β-agonists, reducing airway edema, and suppressing certain proinflammatory cytokines (Respir Med 2004;98:275-84).
   2. Systemic corticosteroids should be administered to patients who have moderate or severe exacerbations or to patients who do not respond promptly and completely to SABA treatment (Am J Med 1983;74:845-51).
   3. Typically, there is a 6–8-hour delay in the response to corticosteroids in status asthmaticus or life threatening asthma; therefore, administration should be considered early in the course (within 1 hour of presentation) (Cochrane Database Syst Rev 2001;1:CD002178).
   4. Oral prednisone is considered as effective as parenteral corticosteroids in most patients; however, it may not be as effective in critically ill patients with impaired gastric absorption.
   5. GINA guidelines recommend prednisone 1 mg/kg/day up to a maximum of prednisone 50 mg daily for 5–7 days for treatment of exacerbations.
      a. Other guidelines recommend longer courses and higher doses for life threatening asthma exacerbations based on the same evidence used by GINA.
      b. For corticosteroid courses less than 2 weeks (GINA 2023), tapering is not necessary, especially if patients are concurrently using an inhaled corticosteroid.
   6. Inhaled corticosteroids can be initiated any time in the treatment of an asthma exacerbation. It is not recommended to replace systemic steroids with inhaled corticosteroids (Respir Care 2018;63:783-96).

N. Adjunctive Therapies (Respir Care 2018;63:783-96; J Clin Med 2019;8:1283)
   1. Ketamine is a phencyclidine derivative that has bronchodilatory properties that reduce airway resistance and prevent the reuptake of norepinephrine, which may stimulate β_2-receptors (Indian J Crit Care Med 2013;17:154-61). Ketamine may be used for as an induction for rapid sequence induction or sedation for MV (Chest 2022;162:747-56).
   2. Helium-oxygen (heliox) is a blended gas (mixture of about 70%–80% helium and 20%–30% oxygen) that decreases airway resistance, leading to improved airflow and ventilation. Use of heliox in status asthmaticus may delay the need for intubation by allowing other therapies to work or as a delivery mechanism for nebulized SABAs.
   3. Magnesium sulfate (2 g intravenously administered over 20–30 minutes) can be considered in patients who have life-threatening exacerbations and are unresponsive to conventional therapies after 1 hour (GINA 2023).
      a. Magnesium is thought to cause bronchodilation by inhibiting calcium channels on smooth muscle, leading to relaxation.
      b. In addition, magnesium may have anti-inflammatory properties that interfere with the activation and release of neutrophils in patients with asthma.
      c. Data for magnesium sulfate administration in severe asthma exacerbations are mixed. In addition, data exist using nebulized magnesium rather than intravenous, but evidence is currently not robust enough to recommend (Front Allergy 2023;4:1211949).
   4. Centers equipped with extracorporeal membrane oxygenation may offer this as a bridge to recovery if a patient's respiratory acidosis cannot be managed by invasive MV alone.

O. Treatments Not Recommended (GINA 2023)
   1. Methylxanthines (theophylline and aminophylline) do not improve lung function or other outcomes in hospitalized adults. Use is associated with severe adverse effects.
   2. Antimicrobials are not generally recommended for acute asthma exacerbations; however, they can be considered if there is evidence of concurrent infection.

Patient Case

Questions 5 and 6 pertain to the following case.
A 30-year-old woman (weight 115 kg) with status asthmaticus is admitted to the ICU. She has a history of severe refractory asthma that has required endotracheal intubation three times in the past 6 months. Her medical history includes hypertension, diabetes, obesity, and bipolar disorder. She has used at least three canisters of albuterol per month for the past 2 months to manage her symptoms.

5. Which best represents the patient's risk factors for higher mortality?
   A. Three hospitalizations in the past 6 months and bipolar disorder.
   B. Use of more than two canisters of SABAs in the past month and obesity.
   C. Hospitalization for asthma in the past month and diabetes.
   D. Prior episode of near-fatal asthma and hypertension.

6. The patient is endotracheally intubated and placed on MV. Which would be the most appropriate initial therapy for this patient with life-threatening asthma exacerbation?
   A. Inhaled albuterol by nebulization 2.5 mg every 4 hours.
   B. Inhaled albuterol by nebulization 2.5 mg every 4 hours and inhaled ipratropium by nebulization 0.5 mg every 6 hours.
   C. Inhaled albuterol by nebulization 2.5 mg every 4 hours, inhaled ipratropium by nebulization 0.5 mg every 6 hours, and methylprednisolone 40 mg intravenously daily.
   D. Inhaled albuterol by nebulization 2.5 mg every 4 hours, inhaled ipratropium by nebulization 0.5 mg every 6 hours, and methylprednisolone 125 mg intravenously every 6 hours.

IV. ACUTE CHRONIC OBSTRUCTIVE PULMONARY DISEASE EXACERBATION

A. Chronic Obstructive Pulmonary Disease (COPD) (GOLD 2024)
   1. Newly defined in the GOLD 2023 guidelines as a heterogeneous lung condition characterized by chronic respiratory symptoms (dyspnea, cough, sputum production, exacerbations) as a result of abnormalities of the airways (bronchitis, bronchiolitis) and/or alveoli (emphysema) that cause persistent, often progressive, airflow obstruction (Table 13).

Table 13. Classification of Airflow Limitation Severity in Chronic Obstructive Pulmonary Disease

Grade	Spirometric GOLD Classification
GOLD 1: Mild	$FEV_1 \geq 80\%$ predicted
GOLD 2: Moderate	$50\% \leq FEV_1 < 80\%$ predicted
GOLD 3: Severe	$30\% \leq FEV_1 < 50\%$ predicted
GOLD 4: Very severe	$FEV_1 < 30\%$ predicted

FEV_1 = forced expiratory volume in the first second of expiration; GOLD = Global Initiative for Chronic Obstructive Lung Disease.

Adapted from: Global Initiative for Chronic Obstructive Lung Disease (GOLD). Global strategy for the diagnosis, management, and prevention of COPD. 2024.

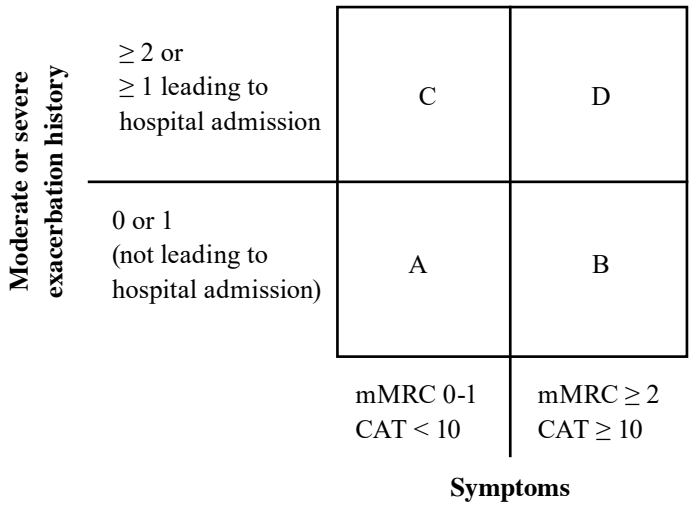


Figure 1. Assessment of Chronic Obstructive Pulmonary Disease Symptoms and Risk of Exacerbations
CAT = Chronic Obstructive Pulmonary Disease Assessment Test; mMRC = Modified British Medical Research Council Questionnaire.

2. Exacerbations of COPD (ECOPD): An event that is characterized by dyspnea and/or cough that worsens over less than 14 days, which may be accompanied by tachypnea and/or tachycardia and is often associated with local and systemic inflammation caused by airway infection, pollution, or other insults to the lungs (Am J Respir Crit Care Med 2021;204:1251-8)
3. Diagnosis of ECOPD
   a. Symptoms, severity of dyspnea using a visual analog scale, and documentation of the presence of cough should be assessed.
   b. Signs (tachypnea, tachycardia), sputum volume and color, and respiratory distress (accessory muscle use) should also be assessed.
   c. Severity should be evaluated with Spo_2, C-reactive protein (CRP), and ABG (venous blood gas to assess pH and bicarbonate is an acceptable alternative).
   d. Identify the etiology of the event (viral, bacterial, or environmental).
   e. Spirometry is not accurate during an exacerbation and is not recommended.
4. Classifications of ECOPD

Table 14. Classification of ECOPD Severity

ECOPD Severity	Description	Diagnostic Parameters
Mild	Treated with short-acting bronchodilators only	Patient must exceed three of the listed variables to move to moderate • Dyspnea VAS < 5 • RR < 24 breaths/min • HR < 95 beats/min • Resting Spo_2 ≥ 92% breathing ambient air (or patient's usual oxygen prescription) AND change ≤ 3%[a] • CRP < 10 mg/L (if available)

Table 14. Classification of ECOPD Severity (*continued*)

ECOPD Severity	Description	Diagnostic Parameters
Moderate	Treated with short-acting bronchodilators and oral corticosteroids with or without the addition of antibiotics	• Dyspnea VAS ≥ 5 • RR ≥ 24 breaths/min • HR ≥ 95 beats/min • Resting Spo_2 < 92% breathing ambient air (or patient's usual oxygen prescription) AND/OR change > 3%[a] • CRP ≥ 10 mg/L (if available) • If ABG obtained, hypoxemia and/or hypercapnia may be present (Pao_2 ≤ 60 mm Hg, $Paco_2$ > 45 mm Hg), but acidosis will be absent
Severe	Patient requires ED visit or hospitalization; patient may also experience acute respiratory failure	• Dyspnea VAS ≥ 5 • RR ≥ 24 breaths/min • HR ≥ 95 beats/min • Resting Spo_2 < 92% breathing ambient air (or patient's usual oxygen prescription) AND/OR change > 3%[a] • CRP ≥ 10 mg/L (if available) • ABG shows new hypercapnia and acidosis ($Paco_2$ > 45 mm Hg and pH < 7.35)

[a]Pulse oximetry (Spo_2) is not as accurate as an ABG, and there is a risk of overestimation of blood oxygenation saturation in patients with darker skin tones (N Engl J Med 2020;383:2477-8).

ABG = arterial blood gas; CRP = C-reactive protein; ECOPD = exacerbations of chronic obstructive pulmonary disease; ED = emergency department; HR = heart rate; RR = respiratory rate; Sao_2 = oxygen saturation; Spo_2 = functional oxygen saturation; VAS = visual analog scale.

Adapted from: The ROME Proposal (Am J Respir Crit Care Med 2021;204:1251-8); GOLD guidelines 2024.

5. Definitions of ECOPD respiratory failure (GOLD 2024)

Table 15. Definitions of ECOPD Respiratory Failure

No respiratory failure	• RR ≤ 24 breaths/min • HR < 95 beats/min • No accessory respiratory muscle use • No change in mental status • Hypoxemia improved with supplemental oxygen by Venturi mask 24%–35% Fio_2 • No increase in $Paco_2$
Acute respiratory failure – non–life threatening	• RR > 24 breaths/min • Accessory respiratory muscle use • No change in mental status • Hypoxemia improved with supplemental oxygen by Venturi mask > 35% Fio_2 • Hypercarbia; $Paco_2$ increased compared with baseline or elevated 50–60 mm Hg
Acute respiratory failure – life threatening	• RR > 24 breaths/min • Accessory respiratory muscle use • Acute changes in mental status • Hypoxemia not improved with supplemental oxygen by Venturi mask > 40% Fio_2 • Hypercarbia; $Paco_2$ increased compared with baseline or elevated > 60 mm Hg or the presence of acidosis (pH ≤ 7.25)

ECOPD = exacerbations of chronic obstructive pulmonary disease; Fio_2 = fraction of inspired oxygen; HR = heart rate; RR = respiratory rate.

B. Common Causes of ECOPD
   1. Respiratory tract infection (40%–50% of ECOPD)
      a. Bacterial (J Infect 2013;67:516-23)
         i. *Streptococcus pneumoniae*, *Haemophilus influenzae*, and *Moraxella catarrhalis* are the most common organisms.
         ii. *Pseudomonas aeruginosa* and other gram-negative organisms are more common in patients with frequent exacerbations (Chest 1998;113:1542-8; Chest 1999;116:40-6; Respir Med 2003;97:770-7; Chest 2007;131:44-52; Eur Respir J 2009;34:1072-8; J Infect 2013;67:516-23).
      b. Viral SARS-CoV-2, influenza, rhinovirus, parainfluenza, respiratory syncytial virus
   2. Medication nonadherence
   3. Temperature change
   4. Air pollution

C. Oxygen
   1. Oxygen therapy is important in managing ECOPD.
   2. Oxygen by nasal cannula or mask should be administered to patients with severe exacerbations to achieve an Sa_{O_2} of 88-92% (BMJ 2010;341:c5462).
   3. Caution is advised with oxygen supplementation in patients with COPD (Crit Care 2012;16:323):
      a. A slight level of hypoxemia may serve as a trigger for their respiratory drive secondary to chronic hypercapnia.
      b. May increase ventilation/perfusion mismatch and decrease respiratory rate centrally and perpetuate the Haldane effect
      c. A recent post hoc analysis of two prospective COPD trials suggests that exceeding an Sa_{O_2} of 92% is associated with increased mortality (HR 1.98 for Sa_{O_2} 93%–96%, HR 2.97 for Sa_{O_2} 97%–100%) (Emerg Med J 2021;38:170-7).

D. High-Flow Oxygen Therapy by Nasal Cannula
   1. Alternative to standard oxygen therapy or noninvasive positive pressure ventilation (GOLD 2024)
   2. Improves oxygenation and ventilation in patients with COPD and very severe underlying disease (Thorax 2016;71:759-61)

E. Noninvasive MV
   1. Indications for noninvasive MV include at least one of the following (GOLD 2024):
      a. Respiratory acidosis (Pa_{CO_2} 45 mm Hg or greater and/or arterial pH 7.35 or less)
      b. Severe dyspnea with clinical signs suggestive of respiratory muscle fatigue, increased work of breathing, or both; use of accessory muscles; paradoxical motion of the abdomen; or retraction of the intercostal spaces
      c. Persistent hypoxemia despite supplemental oxygen
   2. NIV is preferred to invasive MV as the initial management of acute respiratory failure. NIV improves gas exchange, reduces work of breathing and need for MV, decreases hospitalization duration, and improves survival (N Engl J Med 1995;333:817-22; Eur Respir J 2017;49:1600791).

F. Mechanical Ventilation: Indications for invasive MV (GOLD 2024)
   1. Unable to tolerate or failure of noninvasive ventilation
   2. Respiratory or cardiac arrest
   3. Altered level of consciousness
   4. Aspiration or vomiting
   5. Hemodynamic instability despite fluid and vasopressors
   6. Severe ventricular or supraventricular arrhythmias

7. Life-threatening hypoxemia and inability to tolerate noninvasive ventilation
8. Inability to manage respiratory secretions

G. Bronchodilators
1. Inhaled SABAs (nebulized or MDI) with or without a short-acting anticholinergic are preferred for bronchodilation in ECOPD.
2. Continue long-acting β-agonists and anticholinergic agents, although no studies have evaluated this regimen. If patients have not yet started these agents, initiate long-acting agents as soon as patient is stable and prior to discharge.
3. Currently, evidence is also lacking regarding a mode of delivery when comparing nebulizers with MDIs during ECOPD, though continuous nebulization is not recommended (GOLD 2024; Cochrane Database Syst Rev 2016;8:CD011826).
4. Treatments NOT recommended: Methylxanthines (theophylline and aminophylline) because of significant adverse effects

H. Corticosteroid Therapy
1. Corticosteroids shorten recovery time, improve lung function (FEV_1), improve oxygenation, and decrease the risk of early relapse, treatment failure, and length of hospitalization (GOLD 2024; N Engl J Med 1999;340:1941-7; Chest 2001;119:726-30). Prednisone 40 mg daily or equivalent (preferably oral) for 5 days may shorten recovery time in acute ECOPD. If oral administration is not an option, equivalent doses of intravenous methylprednisolone or nebulized budesonide can be administered (GOLD 2024).
2. Literature review
    a. Reduction in the Use of Corticosteroids in Exacerbated COPD (REDUCE) was a randomized, noninferiority trial. Patients (n=314) were randomized to prednisone 40 mg daily for either 5 days or 14 days (inhaled bronchodilators and antimicrobials in both groups). ECOPD occurred in 35.9% of patients in the 5-day group and in 36.8% of patients in the 14-day group (p=0.006) In addition, although one of the secondary endpoints notes no difference between groups regarding the need for mechanical ventilation (defined as either intubation or noninvasive ventilation), it is unclear how many patients in the study population were critically ill. (JAMA 2013;309:2223-31). This study did not include critically ill patients requiring MV.
    b. A randomized, double-blind trial including 83 adult patients with an ECOPD requiring hospitalization and ventilatory support (invasive or noninvasive) compared methylprednisolone 0.5 mg/kg intravenously every 6 hours for 72 hours, 0.5 mg/kg every 12 hours on days 4–6, and 0.5 mg/kg daily on days 7–10 with placebo. Patients receiving corticosteroids had a shorter duration of MV (p=0.04), trend toward shorter length of ICU stay (p=0.09), fewer noninvasive ventilation failures (p=0.004), and more hyperglycemia (p=0.04). This study was not powered to detect differences in length of stay or mortality (Arch Intern Med 2011;171:1939-46).
    c. An open-label, randomized trial included 217 critically ill patients 40 years and older with an ECOPD requiring MV to receive oral prednisone 1 mg/kg daily for a maximum of 10 days or usual care. No differences in ICU mortality, noninvasive ventilation failure, duration of MV, or ICU length of stay were observed. Hyperglycemia significantly increased in the steroid group (*Eur Respir J* 2014;43:717-24).
    d. A cohort study compared high-dose (greater than 240 mg/day) with low-dose (240 mg/day or less) methylprednisolone in more than 17,000 patients with an ECOPD admitted to an ICU. Patients in the high-dose group had longer ICU and hospital lengths of stay, higher hospital costs, longer durations of MV, and more hyperglycemia and fungal infections. Mortality did not differ between groups (Am J Respir Crit Care Med 2014;189:1052-64).

e. An open-label, randomized, parallel-group trial compared fixed-dosed corticosteroid treatment (prednisone 40 mg/day) with variable-dosed corticosteroid treatment for 5 days for the treatment of ECOPD in 246 hospitalized patients. Those in the variable-dosed corticosteroid group received 61.4 mg/day of prednisone on average. In-hospital treatment failure was defined as death, need for MV, administration of additional rescue steroids or aminophylline, or upgrade of antibiotic use. Those in the variable-dosed group experienced a reduced risk of in-hospital treatment failure (RR 0.37 [0.18–0.74]). Post hoc analysis did not show a significant difference in treatment failure for patients receiving less than or greater than 60 mg/day (31.3% vs. 23.4% [p=0.351]); however, treatment failure was higher in patients receiving 40 mg or less per day, 44.4%, than for those receiving greater than 40 mg/day, 22.9% (p=0.27). This study suggests a more appropriate fixed steroid dose for ECOPD is prednisolone 60 mg/day rather than 40 mg/day (Chest 2021;160:1660-9).

I. Antimicrobials (GOLD 2024; Cochrane Database Syst Rev 2018;10:CD010257)
   1. Data are conflicting on which populations benefit from antimicrobials.
      a. A 2018 systematic review comparing antibiotics with placebo for ECOPD found a treatment failure relative risk of 0.72 (n=1191) among studies in outpatient settings using only antibiotics currently in practice. No differences in all-cause mortality or treatment failure were identified (Cochrane Database Syst Rev 2018;10:CD010257).
         i. Only one trial included ICU patients (Lancet 2001;358:2020-5).
            (a) Administration of antibiotics reduced treatment failure, all-cause mortality, and hospital length of stay.
            (b) Study limitations: Conducted in 2001; generalizability to current practice may be limited; study used ofloxacin, which is no longer used
         ii. Data remain inconclusive regarding treatment failure, mortality, and repeated exacerbations for patients who are hospitalized outside the ICU with severe COPD and for outpatients.
         iii. The findings within this systematic review are further limited by the inconsistency among studies in the description of COPD severity at baseline.
   2. GOLD guidelines recommend 5–7 days of antibiotics during ECOPD in the following patient populations:
      a. Patients with all three cardinal symptoms: increase in dyspnea, increased sputum, and sputum purulence
      b. Patients with two cardinal symptoms if purulent sputum is one of the two symptoms
      c. Patients requiring invasive or noninvasive MV for their respiratory insufficiency or failure as the result of an ECOPD
   3. Choice of antimicrobial should be based on local resistance patterns.
      a. Initial therapy with amoxicillin/clavulanate, macrolide, tetracycline, or quinolone is suggested by the guidelines.
         i. Macrolide-based antimicrobial regimens may improve clinical outcomes.
            (a) A propensity score-matched cohort study found that macrolide treatment compared with non-macrolide treatment was associated with reduced 30-day readmission rates (7.3% vs. 8.8%, respectively; p<0.01) (Pharmacotherapy 2019;39:242-52).
      b. For patients with frequent exacerbations, severe airflow obstruction, and/or exacerbations requiring MV, broader-spectrum antimicrobials should be considered because of the possible presence of *P. aeruginosa*.
   4. De-escalation
      a. Antimicrobial therapy should be de-escalated as appropriate according to culture results, if available.

b. Use of procalcitonin in critically ill patients with ECOPD is controversial. A study of ECOPD in ICU patients found that using a procalcitonin-based algorithm for initiating or discontinuing antibiotics was associated with a higher mortality rate compared with patients receiving standard antibiotic regimens (Intensive Care Med 2018;44:428-37).

J. Vitamin D
   1. Mechanism of effect: Vitamin D metabolites attenuate inflammation and support immune system responses to pathogens.
   2. The GOLD guidelines recommend routine identification and supplementation of vitamin D deficiency in patients hospitalized with ECOPD with vitamin D concentrations less than 10 ng/mL.
   3. A systematic review and meta-analysis of three randomized controlled trials (469 patients with data available) showed that supplementation of vitamin D did not affect the rate of moderate/severe ECOPD (Thorax 2019;74:337-45).
      a. Prespecified subgroup analysis of 87 patients identified that vitamin D supplementation reduced the rate of moderate to severe ECOPD in patients whose concentrations were deficient (25-hydroxyvitamin D concentrations less than 10 ng/mL) at baseline.
      b. No effect on exacerbations for patients whose concentrations were not deficient at baseline
   4. In a more recent randomized controlled trial of patients with COPD investigating the effects of vitamin D supplementation on exacerbations, a prespecified subgroup analysis of patients with vitamin D concentrations of 6–10 ng/mL compared with placebo found no difference in time to first exacerbation, time to hospitalization, or rate of exacerbations (Am J Clin Nutr 2022;116:491-9).]

Patient Cases

7. A 79-year-old woman (weight 70 kg) is admitted to the ICU for the management of hypercapnic respiratory failure related to an ECOPD. She has had several admissions for ECOPD. The patient presents with fever, profound dyspnea, increased sputum production (thick and purulent), and confusion. She has a history of anaphylaxis to penicillin. Her blood pressure is 190/100 mm Hg, heart rate is 110 beats/minute, and respiratory rate is 22 breaths/minute. Her chest is hyperinflated and has poor air entry bilaterally. Her ABG values are as follows: pH 7.20, Pco_2 85 mm Hg, and Pao_2 44 mm Hg on 6 L of nasal cannula. She is intubated and placed on MV. Which would be the most appropriate group of medications to treat this patient's severe ECOPD?
   A. Methylprednisolone 1 mg/kg intravenously administered as two divided doses, inhaled albuterol by nebulization, ampicillin/sulbactam 3 g intravenously every 6 hours for 7 days, and azithromycin 500 mg intravenously daily for 5 days.
   B. Prednisone 40 mg by nasogastric tube (NGT) daily for 5 days, inhaled albuterol and ipratropium by nebulization, ampicillin/sulbactam 3 g intravenously every 6 hours for 10 days, and azithromycin 500 mg daily intravenously for 5 days.
   C. Methylprednisolone 1 mg/kg intravenously administered as two divided doses, inhaled albuterol and ipratropium by nebulization, and levofloxacin 750 mg by NGT daily for 7 days.
   D. Prednisone 40 mg by NGT daily for 5 days, inhaled albuterol and ipratropium by nebulization, and levofloxacin 750 mg intravenously daily for 5 days.

8. A 75-year-old woman presents to the ED with a productive cough and altered mental status (oriented only to person). Her medical history is significant for type 2 diabetes and severe COPD. On physical examination, she has a productive cough and is alert and oriented × 4. Her vital signs are temperature 101.8°F, heart rate 95 beats/minute, respiratory rate 33 breaths/minute, systolic blood pressure 105 mm Hg, and Spo_2 86% on her home oxygen of 2 L/minute by nasal cannula. The patient is placed on noninvasive ventilation by Venturi mask at an Fio_2 of 40%, and her Spo_2 is now 91%. A basic metabolic panel returns with SCr 1.3 mg/dL and serum urea nitrogen 55 mg/dL, and an ABG reports pH 7.37, $Paco_2$ 55 mm Hg, and Pao_2 59 mm Hg. Radiography reveals multilobar infiltrates. Which best describes the severity of the patient's ECOPD and her level of respiratory failure according to the 2024 GOLD guidelines?
   A. Moderate ECOPD with non–life-threatening acute respiratory failure.
   B. Moderate ECOPD with life-threatening acute respiratory failure.
   C. Severe ECOPD with life-threatening acute respiratory failure.
   D. Severe ECOPD with non–life-threatening acute respiratory failure.

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ANSWERS AND EXPLANATIONS TO PATIENT CASES

1. **Answer: C**

The patient presents with a CF exacerbation, probably caused by an infection. The most likely causative organism of her infection is *P. aeruginosa*; therefore, therapy must be directed to *P. aeruginosa* (Answer B is incorrect). In addition, antibiotic treatment should include the empiric selection of two antibacterial agents, ideally a β-lactam and an aminoglycoside, dosed to effectively treat the infection (Answers A and D are incorrect). The ideal regimen should include a β-lactam dosed to treat *P. aeruginosa* – in this case, piperacillin/tazobactam at the recommended dose – and an aminoglycoside dosed once daily (Answer C is correct).

2. **Answer: B**

According to the STOP2 trial, patients who improve within 7 days of antimicrobial therapy can complete 10 days of antimicrobials. These patients' clinical outcomes are noninferior to those of patients who continue antimicrobials for 14 days (Answer B is correct). Answer A is incorrect; 7 days is too short for an exacerbation of CF. Answer C is incorrect, given the results of the STOP2 trial. Answer D is incorrect because microbiological cure is neither expected nor likely for patients with CF.

3. **Answer: D**

This patient has signs of right heart failure and rapid progression of symptoms, which place him in the high-risk category for mortality within 1 year (greater than 20%) (Answers A, B, and C are incorrect). In addition, he has symptoms at rest, which place him in WHO FC IV (Answer D is correct).

4. **Answer: A**

The patient's symptoms and physical findings place him in WHO FC IV. His unfavorable response to the vasodilator challenge makes CCBs an undesirable class of medications for him (Answer B is incorrect). Epoprostenol continuous infusion is indicated for patients with PAH presenting with WHO FC IV to improve symptoms, exercise capacity, and hemodynamics. In addition, it is the only treatment shown to reduce mortality in PAH (Answer A is correct). Macitentan and sildenafil could be considered in this patient; however, his elevated liver enzymes do not make macitentan an ideal agent (Answer C is incorrect). Intravenous sildenafil is also not ideal because of the hypotension associated with the intravenous formulation (Answer D is incorrect).

5. **Answer: C**

Risk factors for increased mortality in patients with asthma include (1) history of near-fatal asthma (e.g., requiring mechanical ventilation); (2) hospitalization or ED visit for asthma in the past year; (3) active use of oral corticosteroids or completion of recent course for asthma; (4) not currently using inhaled corticosteroids; (5) use of more than one canister of SABAs per month; (6) poor adherence to inhaled corticosteroid containing asthma medications and/or poor adherence to asthma action plan; (7) social history that includes major psychosocial problems or psychiatric illness; (8) food allergies; (9) comorbidities including pneumonia, diabetes, and arrhythmias. Two of the patient's listed characteristics are risk factors (Answer C is correct). Other answers include one risk factor or none (Answers A, B, and D are incorrect).

6. **Answer: C**

For life-threatening asthma exacerbations, SABAs, short-acting antimuscarinics, and intravenous corticosteroids are recommended (Answers A and B are incorrect). The recommended dose of intravenous corticosteroids is methylprednisolone 40 mg (equivalent to prednisone 50 mg) intravenously per day administered early in the course of the exacerbation (Answer C is correct; Answer D is incorrect).

7. **Answer: D**

The latest GOLD guidelines recommend systemic corticosteroids to shorten recovery time, improve FEV_1, and improve hypoxemia. The recommended dose is prednisone 40 mg orally once daily (or equivalent) for 5 days. Although recent data identified that some patients may have worse outcomes with fixed prednisone 40-mg doses, the data suggest prednisone 60 mg is adequate to balance the benefit of steroids with the risks of adverse effects (Answers A and C are incorrect). Adding inhaled SABAs (nebulized or MDI) with or without a short-acting anticholinergic is the preferred treatment in an ECOPD. Antibiotic treatment for 5–7 days is also indicated because the patient has signs of bacterial infection

(fever, purulent sputum) and a severe ECOPD with respiratory failure. This patient has a history of anaphylactic reaction to penicillin and repeated exacerbations that may predispose her to resistant organisms such as *P. aeruginosa* (Answers A and B are incorrect); therefore, levofloxacin would be the best recommendation (Answer D is correct).

8. **Answer: A**

The current GOLD guidelines provide specific diagnostic parameters to determine the severity of an ECOPD. A respiratory rate greater than 24 breaths/minute and an Spo_2 less than 92% on home oxygen therapy clearly identify this patient as having either a moderate or severe asthma exacerbation. Patients with a moderate ECOPD may have hypercapnia ($Paco_2$ greater than 45 mm Hg) or hypoxia (Pao_2 less than 60 mm Hg); however, they have not yet developed acidosis (pH less than 7.35), whereas those experiencing a severe ECOPD have developed acidosis (Answers C and D are incorrect). Although the degree of respiratory failure is determined using many of the same parameters as the severity of ECOPD, there is a greater focus on patient response to therapy. An improvement in hypoxemia with noninvasive ventilation by Venturi mask with an Fio_2 greater than 35% and the absence of an altered mental status are associated with non–life-threatening respiratory failure (Answer B is incorrect; Answer A is correct).

ANSWERS AND EXPLANATIONS TO SELF-ASSESSMENT QUESTIONS

1. **Answer: D**
It is imperative to recognize that this patient has acute respiratory distress syndrome (ARDS) caused by a CF exacerbation. Therefore, an inclusive therapy plan should include appropriate treatments for ARDS and CF. For ARDS, a lung-protective ventilation strategy (tidal volume 4–6 mL/kg) and a fluid-conservative strategy (CVP less than 4 mm Hg, if not in shock) are of utmost importance (Answers A and C are incorrect because of the CVP goal; Answer D is correct). Appropriate treatment of the CF exacerbation includes empiric therapy for *P. aeruginosa* in the form of optimal doses of a β-lactam and aminoglycoside (Answer B is incorrect because of the inappropriate tobramycin dose and CVP goal of 10–14 mm Hg).

2. **Answer: D**
The highest quality of evidence for improved outcomes with CF exacerbations is with adjunctive therapy for mucous clearance. The guidelines recommend airway clearance options such as aggressive chest physical therapy, nebulized dornase alfa, and hypertonic saline as a grade B recommendation (Answer D is correct). Hypertonic saline would be appropriate, but nebulization of normal saline is not included in any of the recommendations (Answer A is incorrect). Both corticosteroid administration and concurrent administration of intravenous and inhaled antibiotics have a grade I recommendation, meaning evidence is insufficient to support them (Answers B and C are incorrect).

3. **Answer: D**
This patient presents with severe right heart failure. The primary goal is to optimize RV preload by maintaining a net negative fluid balance using gentle diuresis and blood pressure monitoring (Answer D is correct). Dopamine would increase blood pressure; however, it might worsen the patient's tachycardia, thereby worsening her already tenuous clinical status (Answer A is incorrect). Epoprostenol would help decrease pulmonary pressures; however, epoprostenol would potentially worsen the patient's blood pressure because of its peripheral vasodilating effects (Answer B is incorrect). Phenylephrine would not be optimal because this vasopressor might worsen RV function, further elevate pulmonary artery pressure by α1-receptors in the pulmonary vasculature, and potentially induce a reflex bradycardia (Answer C is incorrect).

4. **Answer: C**
Treatment goals for PAH include, but are not limited to, achieving and maintaining WHO FC I or II (Answer A is incorrect), preserving 6MWD to greater than 440 m (Answer B is incorrect), and preserving RV size and function (right atrial pressure less than 8 mm Hg and cardiac index 2.5 L/minute/m^2 or greater) (Answer C is correct). The clinician should normalize BNP to less than 50 ng/L (Answer D is incorrect).

5. **Answer: C**
This patient has shortness of breath at rest that is interfering with his conversational ability, and his FEV_1 is less than 50% of predicted; therefore, his asthma exacerbation would be classified as severe (Answer C is correct). FEV_1 would be greater than 50% in mild/moderate asthma exacerbation (Answers A and B are incorrect). In a life-threatening asthma exacerbation, the patient would have symptoms such as drowsiness, confusion, or silent chest. This patient could progress to life threatening but currently would not be classified as such (Answer D is incorrect).

6. **Answer: B**
This patient presents with near-fatal asthma that is unresponsive to initial therapy. Magnesium sulfate (2 g intravenously administered over 20–30 minutes) can be considered in patients who have life-threatening exacerbations and are unresponsive to conventional therapies after 1 hour (Answer B is correct). Antimicrobials are not routinely recommended for asthma exacerbations if the patient has no evidence of concurrent infection (Answer A is incorrect). Because of the lack of improved outcomes, risk of adverse effects, and superior bronchodilation with SABAs, methylxanthines (theophylline and aminophylline) are not recommended for acute asthma exacerbations (Answer C is incorrect). Mucolytic agents are irritating and may worsen cough and airflow obstruction (Answer D is incorrect).

7. **Answer: B**
The recommended corticosteroid dose for an ECOPD is prednisone 40 mg orally once daily (Answer D is

incorrect; Answer B is correct). Antimicrobial treatment should be initiated if (1) all three cardinal symptoms of an ECOPD (increased dyspnea, increased sputum production, and increased sputum purulence) are present; (2) two of the three cardinal signs are present, with increased sputum purulence as one of the symptoms; or (3) the patient requires noninvasive or invasive ventilation. This patient has no indication for antimicrobials (Answers A and C are incorrect).

8. **Answer: D**

Although the GOLD-recommended dose of steroids for an ECOPD is prednisone 40 mg orally (or intravenously), some studies that included populations of ICU patients have shown higher doses may be needed or may be tolerated with some increase in adverse effects. Currently, doses higher than prednisone 40 mg have not been linked to increased mortality (Answer A is incorrect). A recent investigation identified significant racial bias in the use of peripheral Sa_{O_2} because of frequent overestimation of Sa_{O_2} compared with a direct ABG measurement for patients with darker skin tones. This answer is incorrect because the publication did not investigate patient outcomes. This patient has a higher Sp_{O_2} measure than the patients in the study (92%–96%), who had a high rate of hypoxemia identified on ABG. Answer C is incorrect because no specific dose of macrolide has been linked to mortality, and azithromycin 500 mg intravenously once is the recommended loading dose. Answer D is correct because titrating oxygen therapy to peripheral Sa_{O_2} greater than 88%–92% has been linked to mortality. The correct intervention would have been to decrease the F_{IO_2} when the patient had an Sa_{O_2} greater than 92%.